Information Processing and Retrieval — Project I
Keyword Extraction and Text Summarization

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This report describes an Information Retrieval system for keyword extraction and extractive/abstractive text summarization over the BBC News Summary corpus (2225 documents, five categories). The system combines a hybrid lexical-semantic index with TF-IDF (Lnc.Ltn), BM25, sentence embeddings, Reciprocal Rank Fusion, and Maximal Marginal Relevance. Generative summarization and keyword extraction are performed via two decoder LLMs. The system is evaluated using precision-recall curves, AUPRC, F_β-measure, and embedding-based cosine similarity.

Additional Key Words and Phrases: information retrieval, keyword extraction, text summarization, BM25, TF-IDF, embeddings, MMR, RRF

1 System Design and Motivation

1.1 Overview

This report describes an IR system designed to perform two tasks over the BBC News Summary corpus $D = \{d_1, \ldots, d_{2225}\}$: **keyword extraction** and **text summarization**. The system follows a hybrid architecture combining lexical (sparse) and semantic (dense) representations, and supports both extractive and generative approaches. The pipeline is structured around four core functions: `indexing()`, `keyword_extraction()`, `summarization()`, and `evaluation()`.

1.2 Text Processing Pipeline

A single, consistent preprocessing pipeline is applied to all documents using spaCy (`en_core_web_sm`). For each document, the pipeline produces two parallel representations:

- **Original sentences** — raw sentence segmentation, used as output units for extractive summarization.
- **Processed sentences** — lemmatized tokens (lowercased, stopwords and punctuation removed) combined with noun phrases extracted via spaCy's noun chunking, used for lexical scoring and keyword extraction.

Lemmatization was chosen over stemming for two reasons. First, lemmatization maps words to valid dictionary forms (e.g., *running* → *run*), preserving semantic interpretability. Second, encoder LLMs are trained on natural language text; providing lemmatized but readable tokens ensures that vector representations remain semantically coherent, whereas aggressively stemmed tokens would degrade embedding quality.

Noun phrases (e.g., *prime minister*, *stock market*) are included in the term space alongside individual tokens, enriching both keyword candidates and BM25 scoring.

1.3 Hybrid Index

The index I is hybrid, combining two complementary components.

Lexical component (sparse/inverted index). For each term t in the vocabulary, the inverted index stores the set of documents in which t appears and its frequency per document. This structure enables efficient computation of document frequency df_t, required for TF-IDF and BM25. The average sentence length $\overline{|s|}_c$ is precomputed per category c during indexing to support BM25 normalization.

Semantic component (dense / embeddings). For each document, the index stores one embedding vector per sentence, produced by two lightweight encoder LLMs with distinct profiles:

- `all-MiniLM-L6-v2` — a general-purpose bi-encoder trained with contrastive learning on diverse corpora (*minilm_general*).
- `multi-qa-MiniLM-L6-cos-v1` — fine-tuned specifically for semantic similarity and question-answering tasks (*minilm_qa*).

Sentences are encoded using their **original (raw) text**, not the preprocessed lemmatized version. Encoder LLMs are trained on natural language; feeding them raw text preserves the full linguistic context (grammar, punctuation, capitalization) and produces higher-quality embeddings than feeding lemmatized token sequences.

1.4 Keyword Extraction

Extractive approach (TF-IDF, Lnc.Ltn scheme). Keywords are scored using the Lnc.Ltn TF-IDF weighting scheme. For a term t in document d:

$$\text{score}(t, d) = \frac{(1 + \log_{10} \text{tf}_{t,d})}{\|\mathbf{d}\|} \cdot \log_{10}\left(\frac{N}{df_t}\right) \tag{1}$$

where $\|\mathbf{d}\|$ is the ℓ_2 norm of the document's TF vector (cosine normalization) and $N = 2225$. Candidates are ranked by descending score. A semantic redundancy filter based on cosine similarity between term embeddings (threshold = 0.7) removes near-duplicate keywords while preserving the original TF-IDF ranking order.

Generative approach (prompting). Two lightweight autoregressive decoder LLMs with distinct profiles are prompted to extract keywords:

- `Qwen/Qwen2.5-1.5B-Instruct` — a multi-purpose instruction-tuned causal decoder (1.5B parameters).
- `Qwen3-Reranker-0.6B` — a smaller model oriented towards ranking and retrieval tasks (0.6B parameters).

Two prompt variants are used per model (zero-shot with minimal instructions, and a more directive zero-shot with explicit formatting rules), enabling comparison of prompt sensitivity. A TF-IDF fallback fills missing keywords when the model generates fewer than p outputs.

1.5 Extractive Summarization

Sentence scoring. Three IR models are supported:

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TF-IDF: Score of sentence s is the sum of Lnc.Ltn TF-IDF scores of its terms.

BM25: For sentence s with terms T_s:

$$\mathrm{BM25}(s) = \sum_{t \in T_s} \log\left(\frac{N - df_t + 0.5}{df_t + 0.5} + 1\right) \cdot \frac{\mathrm{tf}_{t,s}(k_1 + 1)}{\mathrm{tf}_{t,s} + k_1\left(1 - b + b \cdot \frac{|s|}{\overline{|s|}_c}\right)} \tag{2}$$

with $k_1 = 1.2$, $b = 0.75$, and $\overline{|s|}_c$ the average sentence length within category c. Using category-level averages provides more appropriate normalization, since articles differ systematically in length across categories.

Embeddings: Score of sentence s is the cosine similarity between its embedding and the document centroid (mean of all sentence embeddings).

Reciprocal Rank Fusion (RRF).. BM25 and embedding rankings are fused via RRF with $\mu = 5$:

$$\mathrm{RRF}(s) = \sum_{f \in \{\mathrm{BM25}, \mathrm{Emb}\}} \frac{1}{\mu + \mathrm{rank}_f(s)} \tag{3}$$

Maximal Marginal Relevance (MMR).. Sentences are selected iteratively to balance relevance and diversity:

$$\mathrm{MMR}(s) = (1 - \lambda) \cdot \mathrm{sim}(s, d) - \lambda \sum_{v \in S} \mathrm{sim}(s, v) \tag{4}$$

where S is the set of already-selected sentences, sim is cosine similarity between sentence embeddings, and $\lambda \in [0, 1]$ controls the diversity-relevance trade-off. Higher λ increases diversity; $\lambda = 0$ recovers greedy relevance-only selection. The final summary preserves the chronological order of selected sentences.

1.6 Generative Summarization

Two pre-trained LLMs are used for abstractive summarization:

- `google/flan-t5-base` — an encoder-decoder model fine-tuned with instruction tuning across diverse NLP tasks.
- `facebook/bart-large-cnn` — a denoising autoencoder fine-tuned specifically on CNN/DailyMail news summarization.

The two models have distinct architectures and pretraining objectives. BART is fed the article text directly (up to 800 characters); Flan-T5 receives an explicit instruction prompt.

1.7 Evaluation

Extractive summaries. A selected sentence is a *hit* if its cosine similarity with any reference sentence exceeds $\tau = 0.5$, computed using `all-MiniLM-L6-v2` as an independent evaluation encoder. This avoids the limitations of exact-match, which systematically underestimates recall. Metrics reported: **Precision-Recall curve** and **AUPRC** (uninterpolated) over all summary lengths $k = 1, \ldots, |d|$; F_β**-measure** at fixed $p = 3$, $\beta = 1$.

Generative summaries. Cosine similarity between the embedding of the generated summary and the reference extract, using the same independent encoder.

2 Question Answering

2.1 Q1. Are keywords extracted under TF-IDF criteria and decoder LLMs in agreement? Why?

TF-IDF and LLM keywords show **partial agreement** in topic but diverge in specificity. Table 1 illustrates this for selected business articles.

Table 1. Keyword comparison for selected BBC business articles.

Document	TF-IDF	LLM (Qwen-1.5B)
business_001	timewarner, catwoman, aol europe	ad_sales, quarterly_profit, time_warner
business_002	greenspan, dollar gains, federal reserve head	dollar, euro, greenspan, trade deficit
business_003	yugansk, rosneft, menatep	loan, rosneft, yukos, yugansk

TF-IDF privileges terms that are *frequent in the document and rare in the collection*, producing highly specific, low-frequency terms (e.g., *catwoman*, *menatep*) that are discriminative but may lack topical generality. LLMs identify *conceptually central* terms (e.g., *trade deficit*, *quarterly profit*) that are more informative to a human reader but less statistically discriminative.

The divergence arises from the fundamentally different nature of the two approaches: TF-IDF is a statistical signal grounded in term rarity, while LLMs leverage distributional semantics and world knowledge. The two approaches are therefore complementary rather than equivalent.

2.2 Q2. How does the extractive summarization system perform for the full collection and within each category?

Table 2 reports F_1 and AUPRC per category (BM25, $\lambda = 0.5$, $p = 3$, 4 documents per category).

Table 2. Per-category extractive summarization performance (BM25, $p = 3$, $\beta = 1$).

Category	F_1	σ_{F_1}	AUPRC	σ_{AUPRC}
Business	0.619	0.332	0.420	0.108
Entertainment	0.733	0.212	0.549	0.066
Politics	0.785	0.133	0.491	0.264
Sport	0.632	0.191	0.400	0.130
Tech	0.760	0.080	0.488	0.155
Overall	**0.706**	0.218	**0.465**	0.171

Politics and tech articles achieve the highest F_1 scores (0.785 and 0.760 respectively), while business and sport score lower. Two factors may explain this: (1) **structural coherence** — political and technical articles tend to have a dominant topic, making the most relevant sentences easier to identify; (2) **lexical specificity** — sports and business articles contain many named entities and statistics

with high TF but moderate IDF, reducing discriminative power. Tech articles also show the lowest variability ($\sigma = 0.08$), suggesting more consistent performance on technical content.

2.3 Q3. How do IR models affect summaries? Are dense representations more effective than sparse ones?

Figure 1 shows precision-recall curves for BM25, embeddings, and RRF.

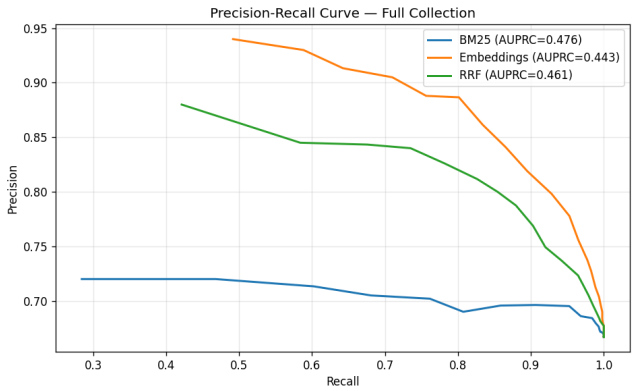

Fig. 1. Precision-Recall curves for BM25, Embeddings, and RRF (100 documents).

Embeddings achieve substantially higher precision at low recall (≈0.94 vs ≈0.72 for BM25), confirming that dense representations are more effective at identifying the most relevant sentences. However, BM25 achieves the highest AUPRC (0.476 vs 0.443), because its precision curve is remarkably flat — it degrades less as more sentences are selected. Table 3 compares all configurations including the QA-tuned encoder.

Table 3. IR model comparison ($p = 3$, $\lambda = 0.5$, 20 documents).

Configuration	F_1	**AUPRC**
BM25 (no MMR)	0.791	0.235
BM25 + MMR	0.616	0.202
Embeddings (general)	0.616	0.156
RRF (general enc.)	0.616	0.174
Embeddings (QA)	0.595	0.143
RRF (QA enc.)	0.595	0.151

The general-purpose encoder outperforms the QA-tuned encoder across all metrics. This is consistent with the nature of the BBC corpus: news articles are general-domain text, and the general encoder is better calibrated for semantic similarity in this domain than a model fine-tuned on question-answering pairs.

2.4 Q4. How does λ impact summarization accuracy? Should λ be fixed or document-specific?

Figure 2 shows the impact of varying λ averaged over 10 randomly selected documents.

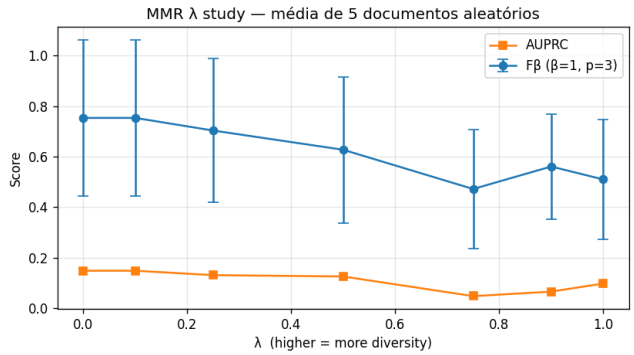

Fig. 2. Impact of λ on summarization quality (mean ± std, 10 documents).

F_1 decreases monotonically from 0.75 ($\lambda = 0$) to 0.51 ($\lambda = 1$), indicating that for this corpus **relevance is more important than diversity** when evaluated against reference extracts. This is partly explained by the nature of the reference summaries, which were themselves generated by a relevance-oriented extractive method.

However, the error bars are large ($\sigma \approx 0.35$), indicating high variability across documents. This strongly suggests that the optimal λ is **document-specific**: a thematically coherent document benefits from low λ, while a redundant document benefits from higher λ. A fixed global threshold is therefore suboptimal; an adaptive λ depending on the document's internal redundancy would be more appropriate.

2.5 Q5. Under embedding-based quality stances, is abstractive summarization superior to extractive summarization?

Table 4 compares extractive and generative summaries using cosine similarity with the reference extract.

Table 4. Cosine similarity with reference extract (embedding-based evaluation).

Method	**Cosine Sim**	σ
Extractive (BM25 + MMR)	0.781	0.106
Generative (`flan-t5-base`)	0.760	0.0992
Generative (`bart-large-cnn`)	0.725	0.104

Contrary to common assumption, the extractive system achieves the **highest semantic similarity with the reference** (0.781 vs 0.760 and 0.725). Two explanations are offered: (1) **Reference bias** — the BBC reference summaries are themselves extractive, so extractive systems naturally share vocabulary and phrasing with both the article and the reference; (2) **Model size** — the generative models used are lightweight, and their abstractive outputs may paraphrase content in ways that reduce embedding similarity despite being semantically equivalent. `flan-t5-base` outperforms `bart-large-cnn` (0.760 vs 0.725), likely due to its broader instruction tuning across diverse NLP tasks, which produces more

controlled and focused summaries even outside its primary training domain. Both generative models remain below the extractive baseline.

Note on prompt variants. Each generative model was evaluated using a zero-shot prompt with minimal instructions (`prompt_a`), applied consistently across all 20 documents (4 per category).

Note on generative evaluation. The generative summaries were produced using two prompt variants per model (zero-shot with minimal instructions, and zero-shot with explicit formatting rules). For evaluation purposes, the zero-shot prompt (`prompt_a`) was used consistently across all documents. Results reported correspond to 20 documents (4 per category).

2.6 Q6. Is Reciprocal Rank Fusion (RRF) useful to aid decisions?

From Table 3, RRF (general encoder) achieves $F_1 = 0.616$ and AUPRC $= 0.174$, identical in F_1 to embeddings alone but with a higher AUPRC. This suggests that RRF provides modest benefit in ranking quality across all summary lengths, but does not improve precision at the fixed length $p = 3$.

The limited gain from RRF can be explained by the asymmetry between the two constituent models: BM25 has substantially lower precision than embeddings. When one model is significantly weaker, the consensus is pulled towards weaker decisions, reducing the benefit of fusion. RRF would be more beneficial if the two models had comparable and complementary strengths.

In conclusion, **RRF is marginally useful** in this setting — it improves AUPRC slightly but does not significantly outperform the best individual model (embeddings). Its utility is expected to increase when combining models of similar quality with diverse error profiles.